

Question 01

$(-5, 2)$ to $(25, 40)$

$$\textcircled{a} \quad P(t) = P_0 + t(P_1 - P_0)$$

$$= (x_0, y_0) + t(x_1 - x_0, y_1 - y_0)$$

$$= (-5, 2) + t(25+5, 40-2)$$

$$= (-5, 2) + t(30, 38)$$

$$= (-5+30t, 2+38t)$$

$$P\left(\frac{2}{3}\right) = \left(-5 + 30 \times \frac{2}{3}, 2 + 38 \times \frac{2}{3}\right)$$

$$= (15, 27.\overline{33})$$

We know points will be inside the line segment only if

$$0 \leq t \leq 1$$

Since, $t = 5 > 1$, the point corresponding to $t = 5$ lies outside the line segment.

$$P(5) = (-5 + 30 \times 5, 2 + 38 \times 5)$$

$$= (145, 192)$$

⑥ The parametric equation helps the cyrus-beck algorithm figure out exactly where a line crosses the edges of a clipping window. Each edge of the convex clipping window has a normal vector, which helps classify intersections as entering and exiting the region. By computing the values of "t" where the line crosses these boundaries, we determine the clipped segment.

$$\textcircled{a} \quad \begin{array}{ll} X_{\min} = -15 & Y_{\min} = 5 \\ X_{\max} = 60 & Y_{\max} = 120 \end{array}$$

$$(x_0, y_0) = (20, 30); (x_1, y_1) = (90, 80)$$

$$\begin{aligned} D &= (x_1 - x_0, y_1 - y_0) \\ &= (90 - 20, 80 - 30) \\ &= (70, 50) \end{aligned}$$

Initially, $t_E = 0$ and $t_L = 1$

Boundary	N_i	$N_i \cdot D$	t	PE/PL	t_E	t_L
Left	$(-1, 0)$	-70	$-\frac{20+15}{90-20} = -\frac{1}{2}$	PE	0	1
Right	$(1, 0)$	70	$-\frac{20-60}{90-20} = \frac{4}{7}$	PL	0	0.57
Bottom	$(0, -1)$	-50	$-\frac{30-5}{80-30} = \frac{1}{2}$	PE	0	0.57
Top	$(0, 1)$	50	$-\frac{30-120}{80-30} = \frac{9}{5}$	PL	0	0.57

$$P(0) = (20, 30)$$

$$P(0.57) = (x_1, y_1) + 0.57(70, 50)$$

$$= (90, 80) + (39.9, 28.5)$$

$$= (129.9, 108.5)$$

Question 02

@ Cohen-Sutherland fails:

① If the clipping region is non rectangular.

② If the region is 3D.

Drawbacks

- * unnecessary clipping perform extra calculations.
- * order of clipping affects efficiency.
- * Repeated outcode computations.

$$\textcircled{b} \quad X_{\min} = -40 \quad Y_{\min} = -20$$
$$X_{\max} = 20 \quad Y_{\max} = 30$$

$$(X_1, Y_1) = (-10, -50); (X_2, Y_2) = (15, 25)$$

oc1 for (X_1, Y_1)

$$X_1 > X_{\min}; L=0$$

$$X_1 < X_{\max}; R=0$$

$$Y_1 < Y_{\min}; B=1$$

$$Y_1 < Y_{\max}; A=0$$

$$\text{oc1} = 0100$$

oc2 for (X_2, Y_2)

$$X_2 > X_{\min}; L=0$$

$$X_2 < X_{\max}; R=0$$

$$Y_2 > Y_{\min}; B=0$$

$$Y_2 < Y_{\max}; A=0$$

$$\text{oc2} = 0000$$

OC1 AND OC2

$$\begin{array}{r} 0100 \\ 0000 \\ \hline 0000 \end{array}$$

Bottom intersection,

$$y = y_{\min} = -20$$

$$x = x_1 + \frac{1}{m}(y_{\min} - y)$$

$$= -10 + \frac{1}{3}(-20 + 50)$$

$$= 0$$

$$\text{new } (x_1, y_1) = (0, -20)$$

new OC1 for new (x_1, y_1)

$$x_1 > x_{\min}; L = 0$$

$$x_1 < x_{\max}; R = 0$$

$$y_1 = y_{\min}; B = 0$$

$$y_1 < y_{\max}; A = 0$$

$$OC1 = 0000$$

$OC1 = 0000$; $OC2 = 0000$; so, completely inside.

segment $(0, -20)$ to $(15, 25)$

$$\left| \begin{array}{l} m = \frac{25+50}{15+10} = 3 \\ \frac{1}{m} = \frac{1}{3} \end{array} \right.$$